MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)
MG5DSEMAT300: DIFFERENTIAL EQUATIONS AND APPLICATIONS

Duration: 2 hrs Maximum Marks: 70

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Answer any five questions. Each question carries 2 marks

1. Define exact differential equations. [K][1]
2. Define Bernoulli differential equation [K][1]
3. What is mean by orthogonal trajectory of a family of curves. [K][2]
4. What is the integrating factor of the differential equation $M(x,y)dx + N(x,y)dy = 0$. [U][1]
5. State the basic existence theorem. [K][3]
6. What is the auxiliary equation of the nth order linear differential equation [U][3]
7. State Sonia Kowulewski's theorem [K][4]
8. State the theorem which gives the method of solving linear equations of the form $Pp + Qq = R$. [K][4]

Part B
Answer any five questions. Each question carries 6 marks

9. Check whether the differential equation $(2xSiny+y^3e^x)dx+(x^2Cosy+3y^2e^x)dy = 0$ is exact or not. [A][1]
10. Solve $\frac{dy}{dx} + (\frac{2x+1}{x})y = e^{-3x}$ [A][1]
11. Solve $(x + 2y + 3)dx + (2x + 4y - 1)dy = 0$ [A][1]
12. Find the orthogonal trajectories of the family of parabolas $y = cx^2$. [A][1]
13. Show that the solutions $sinx$ and $cosx$ of $\frac{d^2y}{dx^2} + y = 0$ are linearly independent. [U][3]

14. Given that $y = x$ is a solution of $(x^2+1)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 0$. Find a linearly independent solution by reducing the order. [A][3]

15. Find the integral curve of the equations $\frac{dx}{x+z} = \frac{dy}{y} = \frac{dz}{z+y^2}$. [A][4]

16. Find the general solution of the differential equation $x^2\frac{\partial z}{\partial x} + y^2\frac{\partial z}{\partial y} = (x+y)z$. [A][4]

Part C
Answer any three questions.
Each question carries 10 marks

17. Solve the initial value problem $(2xCosy + 3x^2y)dx + (x^3 - x^2Siny - y)dy = 0$, $y(0) = 2$. [A][1]

18. (i)Find the curve for which the normal makes equal angles with the radius vector and the initial line. [A][2]
(ii)Find the shape of a reflector such that light coming from a fixed source is reflected in perallel rays. [A][2]

19. Solve the following differential equation using a special transformation $(x - 2y + 1)dx + (4y - 3y - 6)dy = 0$ [A][1]

20. Solve $(x^2+1)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 6(x^2+1)^2$. [A][3]

21. Solve the equation $\frac{dx}{y+\aleph z} = \frac{dy}{z+\beta x} = \frac{dz}{x+\gamma y}$ [A][4]

22. (i) If u is a function of x, y and z which satisfies the PDE $(y-z)\frac{\partial u}{\partial x} + (z-x)\frac{\partial u}{\partial y} + (x-y)\frac{\partial u}{\partial z} = 0$. Show that u contains x, y and z only in combinations $x+y+z$ and $x^2+y^2+z^2$. [A][4]
(ii) Find the general integrals of the linear PDE $z(xp - yq) = y^2 - x^2$ [A][4]